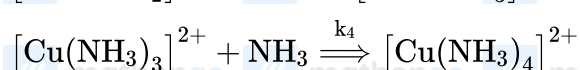
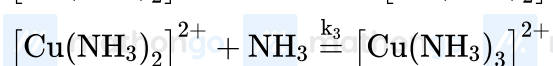
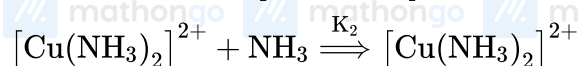
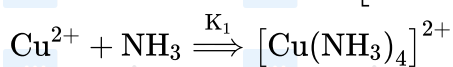


Q1: 24 Feb (Shift 1) - Numerical

The stepwise formation of $[\text{Cu}(\text{NH}_3)_4]^{2+}$ is given below:



The value of stability constants K_1 , K_2 , K_3 and K_4 are 10^4 , 1.58×10^2 , 5×10^2 and 10^2

respectively. The overall equilibrium constants for dissociation of $[\text{Cu}(\text{NH}_3)_4]^{2+}$ is $x \times 10^{-12}$. The value of x is _____. (Rounded off to the nearest integer)

Q2: 24 Feb (Shift 1) - Numerical

At 1990 K and 1 atm pressure, there are equal number of Cl_2 molecules and Cl atoms in the reaction mixture.

The value of K_p for the reaction $\text{Cl}_{2(g)} = 2\text{Cl}_{(g)}$ under the above conditions is $x \times 10^{-1}$. The value of x is _____. (Rounded off to the nearest integer)

Q3: 24 Feb (Shift 1) - Numerical

For the reaction $\text{A}_{(g)} \rightarrow \text{B}_{(g)}$, the value of the equilibrium constant at 300 K and 1 atm is equal to 100.0. The

value of $\Delta_r G$ for the reaction at 300 K and 1 atm in J mol^{-1} is $-xR$, where x is _____. (Rounded off to the nearest integer) [$R = 8.31 \text{ J mol}^{-1} \text{ K}^{-1}$ and $\ln 10 = 2.3$]

Q4: 25 Feb (Shift 1) - Numerical

The ionization enthalpy of Na^+ formation from $\text{Na}_{(g)}$ is $495.8 \text{ kJ mol}^{-1}$, while the electron gain enthalpy of Br is $-325.0 \text{ kJ mol}^{-1}$. Given the lattice enthalpy of NaBr is $-728.4 \text{ kJ mol}^{-1}$. The energy for the

formation of NaBr ionic solid is (-) _____ $\times 10^{-1} \text{ kJ mol}^{-1}$.

Q5: 26 Feb (Shift 1) - Numerical

A homogeneous ideal gaseous reaction $AB_{2(g)} \rightleftharpoons A_{(g)} + 2 B_{(g)}$ is carried out in a 25 litre flask at 27°C . The initial amount of AB_2 was 1 mole and the equilibrium pressure was 1.9 atm. The value of K_p is $x \times 10^{-2}$. The value of x is ____ $[R = 0.08206 \text{ dm}^3 \text{ atm K}^{-1} \text{ mol}^{-1}]$

Q6: 26 Feb (Shift 2) - Numerical

The average S-F bond energy in kJ mol^{-1} of SF_6 is ____ (Rounded off to the nearest integer)

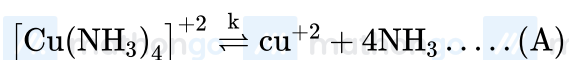
[Given : The values of standard enthalpy of formation of $\text{SF}_6(\text{g})$, $\text{S}(\text{g})$ and $\text{F}(\text{g})$ are -1100, 275 and 80 kJ mol^{-1} respectively.]

Answer Key**Q1 (1)****Q2 (5)****Q3 (1380)****Q4 (5576)****Q5 (74)****Q6 (309)**

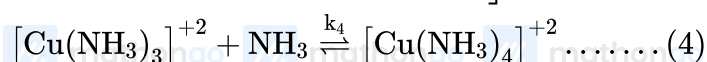
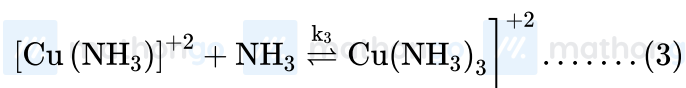
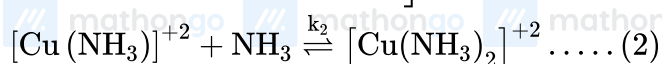
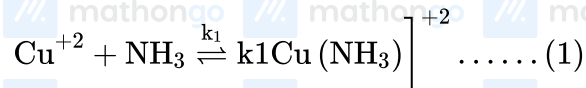
Hints and Solutions

MathonGo

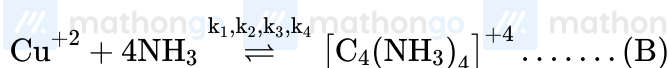
Q1 (1)



For this :



$$(1) + (2) + (3) + (4)$$



So for (A)

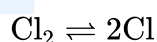
$$K = \frac{1}{k_1 K_2 \cdot K_3 \cdot K_4}$$

Putting the value of k_1 , k_2 , k_3 and k_4 .

$$K = \frac{1}{(10)^4 \cdot (1.58 \times 10^3) (5 \times 10^2) (10)^2} = 1.26 \times 10^{-12}$$

$$x = 1$$

Q2 (5)



Let's mole of eq.

P.P. at eq.

$$\begin{array}{ccc} x & & x \\ \frac{x}{2x} \times 1 & & \frac{x}{2x} \times 1 \\ \frac{1}{2} & & \frac{1}{2} \end{array}$$

$$K_p = \frac{[\text{P}_{\text{Cl}}]^2}{[\text{P}_{\text{Cl}_2}]} = \frac{\left[\frac{1}{2}\right]^2}{\frac{1}{2}} = \frac{1}{2} = 0.5 = 5 \times 10^{-1}$$

$$X = 5$$

Q3 (1380)

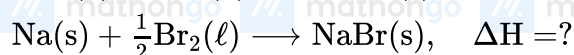
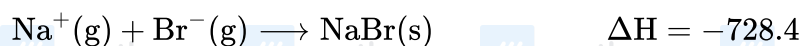
$$\Delta G^\circ = -RT \ln K_{\text{eq}}$$

$$= -R \times 300 \times \ln(10^2)$$

$$= 300 \times 2 \times 2.3 \times (-R)$$

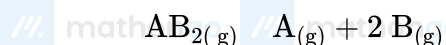
$$= -1380R$$

$$x = 1380 \text{ ans.}$$

Q4 (5576)

$$\Delta H = 495.8 - 325 - 728.4$$

$$-557.6 \text{ kJ} = -5576 \times 10^{-1} \text{ kJ}$$

Q5 (74)

$$\begin{array}{ccc} \text{initial} & 1-x & x \quad 2x \end{array}$$

$$\text{at eq.} \quad \frac{1}{1+2x} \quad \frac{1}{1.9}$$

$$\text{By ratio of pressure \& mole } \frac{1}{1+2x} = \frac{0.985}{1.9}$$

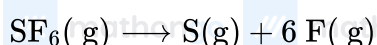
$$1.9 = 0.985 + 1.9x$$

$$0.915 = 1.9x$$

$$\frac{0.915}{1.9} = x; \quad K_p = \frac{4x^2 \cdot x}{(1-x)} \left[\frac{P_{\text{total}}}{n_{\text{total}}} \right]^2$$

$$\Rightarrow \frac{4x^3}{1-x} \left(\frac{RT}{V} \right)^2$$

$$\text{On substituting the values } K_p = 74 \times 10^{-2}$$

Q6 (309)

$$\Delta H_{\text{reaction}}^\circ = 6 \times E_{\text{S-F}} = \Delta H_f^\circ[\text{S}(\text{g})] + 6 \times \Delta H_f^\circ[\text{F}(\text{g})] - \Delta H_f^\circ[\text{SF}_6(\text{g})]$$

$$6 \times E_{\text{S-F}} = 275 + 6 \times 80 - (-1100)$$

$$= 275 + 480 + 1100$$

$$6 \times E_{\text{S-F}} = 1855$$

$$E_{\text{S-F}} = \frac{1855}{6} = 309.1667$$

$$\simeq 309 \text{ kJ/mol Ans.}$$

